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**IBM BOB 2.0**  
**HACKATHON**

# BREAKPOINT AI

Autonomous DevSecOps & Red/Blue/Judge Fuzzing Platform

Built with IBM Bob 2.0

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Powered by IBM watsonx.ai & Meta Llama-3-3-70B-Instruct

# THE PROBLEM

## EXPLOSIVE OUTPUT VS. HIDDEN VULNERABILITIES

AI coding assistants allow developers to ship 10× faster, but introduce subtle business logic flaws, race conditions, and injection vulnerabilities at scale.

## STATIC LINTERS ARE BLIND TO CONTEXT

Traditional SAST tools only match basic patterns—they cannot reason about API contracts, authentication bypasses, or asynchronous state mutation.

## MANUAL SECURITY REVIEWS CREATE RELEASE BOTTLENECKS

DevSecOps teams cannot keep up with thousands of AI-generated lines, leading to unvetted PR merges or delayed release cycles.

# THE SOLUTION

## BreakPoint AI

### ADVERSARIAL FUZZING LOOP (RED / BLUE / JUDGE)

Moves beyond passive code summarization by actively attacking code, writing minimal surgical patches, and verifying fixes against real test suites.

### OFFLINE POLYGLOT STATIC ENGINE

Features 60+ built-in security rules across 16 programming languages (JS/TS, Python, C/C++, Java, Go, Rust, PHP, Swift, Kotlin, etc.) running with zero API token cost.

### DECOUPLED & UNIVERSAL INTEGRATION

Operates anywhere: a global CLI (breakpoint), a stdio MCP server for AI IDEs, a real-time Web C2 Dashboard (localhost:4242), and a headless CI runner.

# DEEP DIVE

## THE RED / BLUE / JUDGE ADVERSARIAL LOOP

● Red Team Agent (Attacker Persona)

● Blue Team Agent (Defender Persona)



Judge Agent (Veto Authority)

# PROTOTYPE TERMINAL SNAPSHOT

```
Windows PowerShell  X + v

[08:53:48] The Blue Team's patches did not address the vulnerabilities identified by the Red Team, and the system remains vulnerable to concurrency attacks,
SQL injection, and insufficient balance checks. New concerns include potential XSS and CSRF attacks.
[08:53:48]
❌ REJECTED - proceeding to round 5...

[08:53:48]
=== Starting Round 5 of 5 ===

ROUND 5 - RED TEAM

[08:54:04]    ↳ Red Team response received
[08:54:04]    Vulnerabilities found: 3

ROUND 5 - BLUE TEAM

[08:54:23]    ↳ Blue Team response received
[08:54:23]    Patches applied: 3
[08:54:23]    ↳ Patch applied → D:\Aryav\Hackathons\IBM Bob 2.0\breakpoint-ai\demo-app-2\routes\transfer.js
[08:54:23]    ↳ Running npm test in D:\Aryav\Hackathons\IBM Bob 2.0\breakpoint-ai\demo-app-2
[08:54:26]    ↳ Test run complete (exit code 1)

ROUND 5 - JUDGE

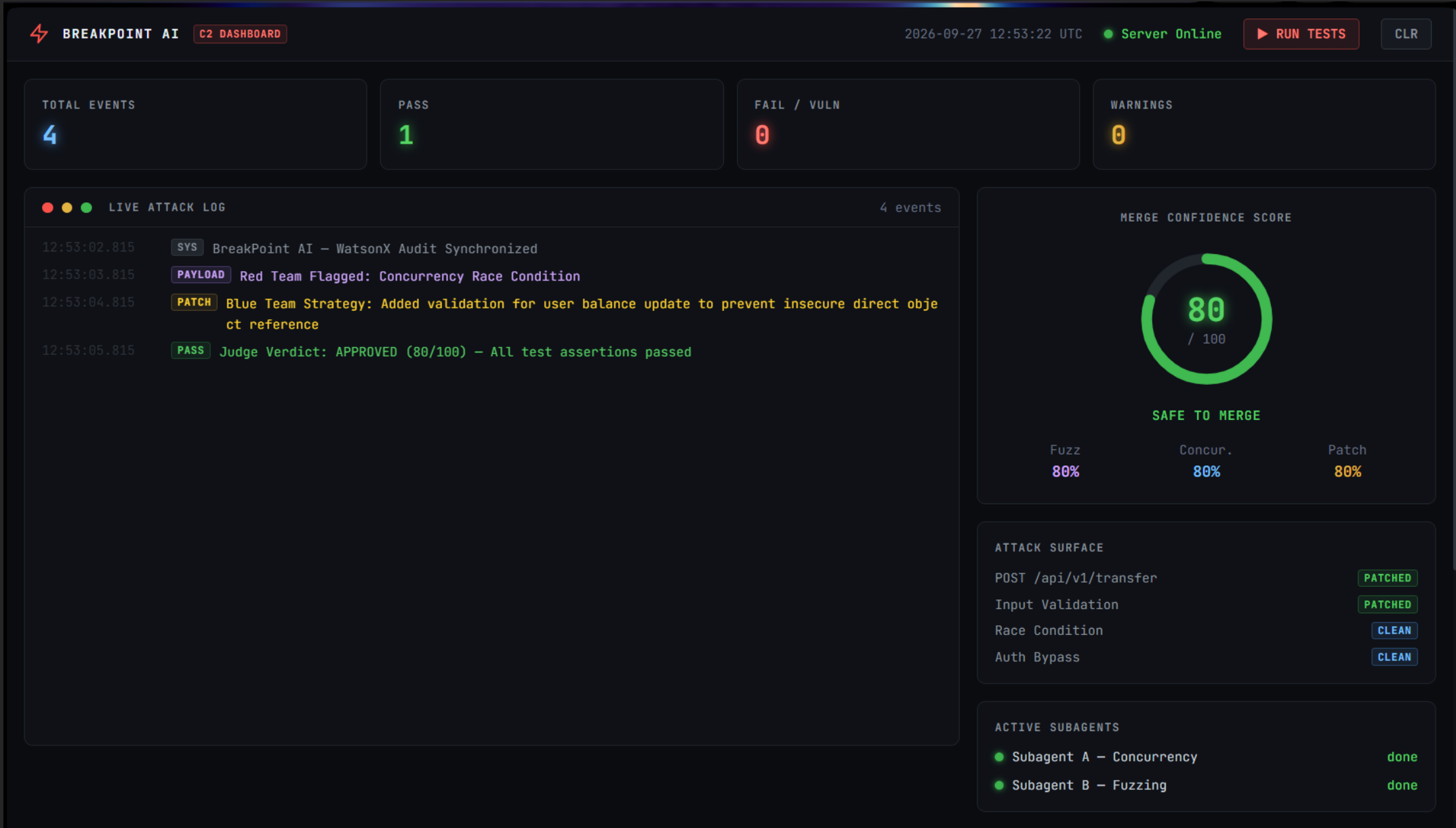
[08:54:30]    ↳ Judge response received
[08:54:30]    ↳ Report saved → D:\Aryav\Hackathons\IBM Bob 2.0\breakpoint-ai\audit-reports\round-5.json

ROUND 5 VERDICT: APPROVED (score: 80/100)

[08:54:30] The Blue Team has successfully patched the identified vulnerabilities, and the updated code includes enhanced input validation and improved error
handling. However, additional security measures are recommended to further prevent abuse and denial of service attacks.
[08:54:30]
✅ APPROVED - all vulnerabilities resolved. Exiting cleanly.

[08:54:30]
↳ Final audit report → D:\Aryav\Hackathons\IBM Bob 2.0\breakpoint-ai\audit-reports\final-audit.json
```

# PROTOTYPE DASHBOARD SNAPSHOT



# TECHNICAL ARCHITECTURE & CONTEXT MANAGEMENT

## Scalable Monorepo Architecture

- `cli/`: Command routing, auto-discovery engine, and static AST/regex rules.
- `dashboard/`: Express server broadcasting Server-Sent Events (SSE) to a React/Tailwind frontend.
- `mcp-server/`: Stdio server exposing `scan_ast`, `execute_fuzzer`, and `apply_patch` to AI assistants.

## Windowed Context Budget Engine

- Solves LLM context truncation on large codebases.
- Dynamically scores files, prioritizes source code with static findings, and enforces a `SOURCE_BUDGET_CHARS = 16,000` window with line-range excerpts for Blue Team patches.

# IBM BOB 2.0 & IBM WATSONX.AI INTEGRATION

## IBM BOB 2.0 (DEVELOPER FORCE MULTIPLIER)

Used throughout development to scaffold the monorepo, author multi-language security rules, write React custom streaming hooks (useSSE.js), and configure build tools. Created reusable project configuration templates (breakpoint.config.json) and automated CI scripts (ci-fuzzer.js).

## IBM WATSONX.AI (INFERENCE BACKBONE)

Authenticates via IBM Cloud IAM (iam.cloud.ibm.com) using API keys to communicate directly with watsonx text chat APIs. Powers the Red, Blue, and Judge reasoning personas using the IBM Granite 3-3-8B Instruct foundation model.

# BUSINESS IMPACT, DEVSECOPS VALUE & FUTURE ROADMAP

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## Enterprise DevSecOps Value

- Shift-Left Security: Catches OWASP Top 10 vulnerabilities and logic bugs during local dev or CI, reducing post-release breach remediation costs by up to 80%.
- Zero Token Cost Baseline: Offline static scans run instantly locally; expensive LLM calls are reserved exclusively for dynamic fuzzing and patching.

## Future Roadmap

- Containerized MicroVM Sandboxing: Isolating dynamic payload execution inside Firecracker microVMs.
- Automated Pull Request Remediation: Opening verified, test-backed fix PRs automatically in GitHub Actions.

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**THANK YOU**